Introduction to Text Mining

By:-

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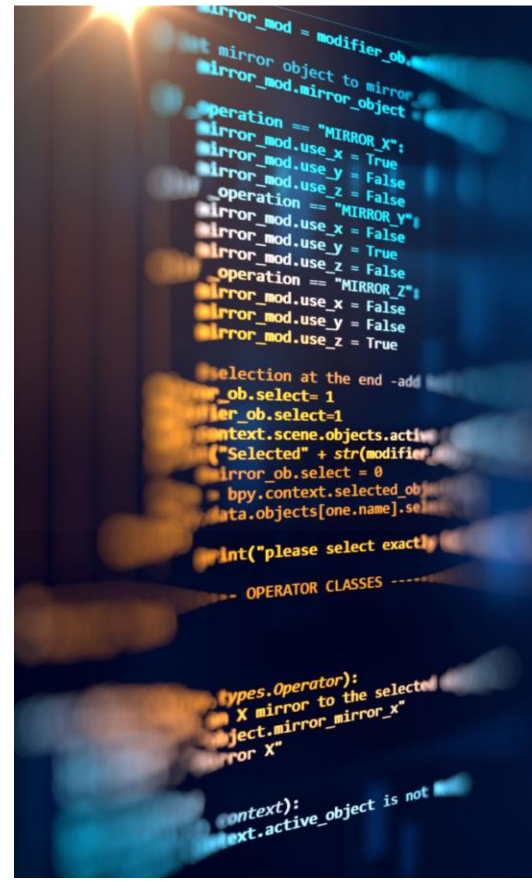
Definition of Text Mining

- Text mining is a knowledge-intensive process for extracting useful information from unstructured textual data.

- It combines techniques from Natural Language Processing (NLP), machine learning, and data mining to analyze textual information.

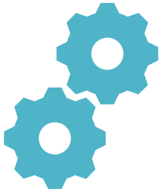
- The goal is to transform raw text into structured data for tasks such as classification, clustering, sentiment analysis, and trend detection.

- Text mining helps organizations uncover hidden information, support decision-making, and automate understanding of vast text sources.



Data Mining versus Text Mining

Both seek for novel and useful patterns



Both are semi-automated processes

Difference is the nature of the data

Data Mining versus Text Mining

Structured versus unstructured data

Structured data: in databases

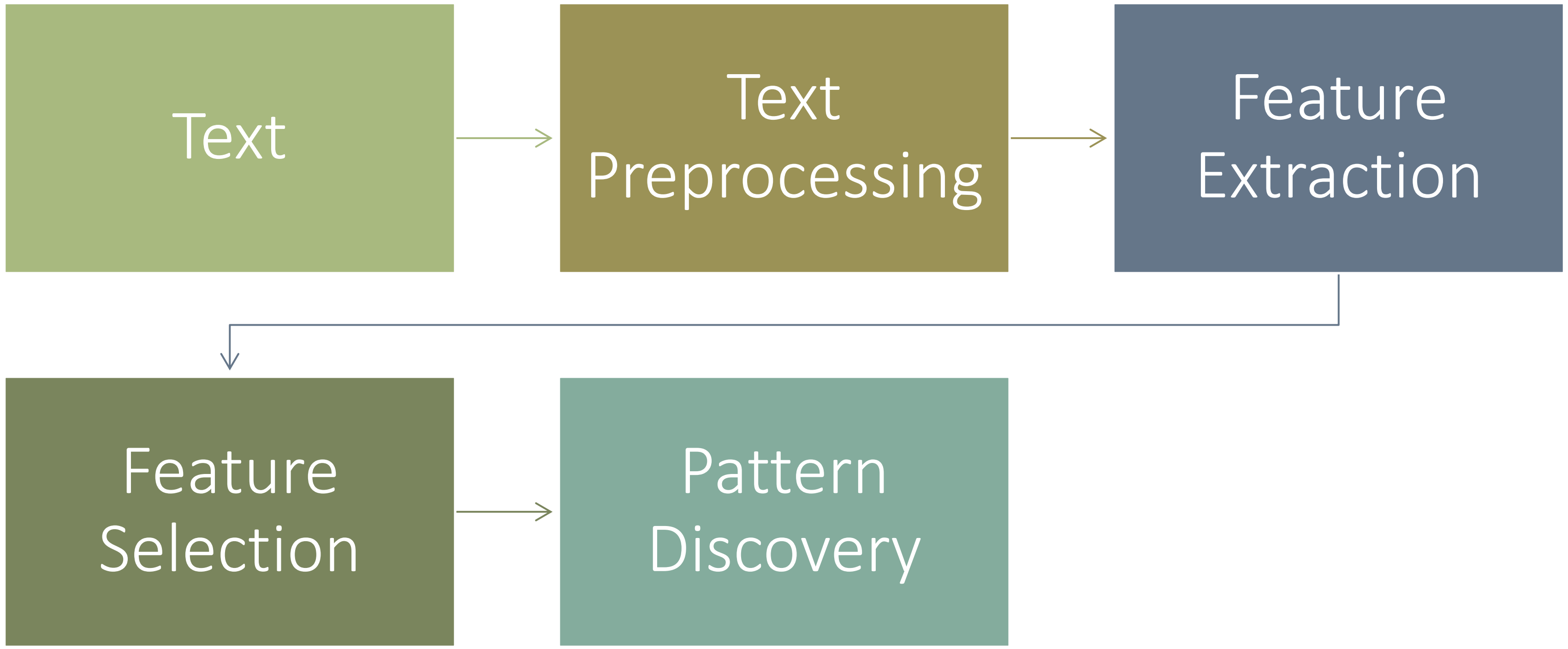
Unstructured data: Word documents, PDF files, text excerpts, XML files, and so on

To perform text mining – first, impose structure to the data, then mine the structured data.

Why Text Mining?

- Roughly 85-90 percent of all corporate data is in some kind of unstructured form (e.g., text)
- Unstructured corporate data is doubling in size every 18 months…
- Tapping into these information sources is not an option, but a need to stay competitive
- A semi-automated process of extracting information and discovering knowledge from unstructured data source

Text Mining Process



Text Preprocessing

- Transforms unstructured data into a structured format suitable for analysis.
- include all those routines, processes, and methods required to prepare data for a text mining system's core knowledge discovery operations.
- Includes routines to prepare data for core knowledge discovery operations.

Feature Extraction

Extracts key elements like

Words

terms

Concepts

In order to represent documents.

Feature Selection

Pick important features

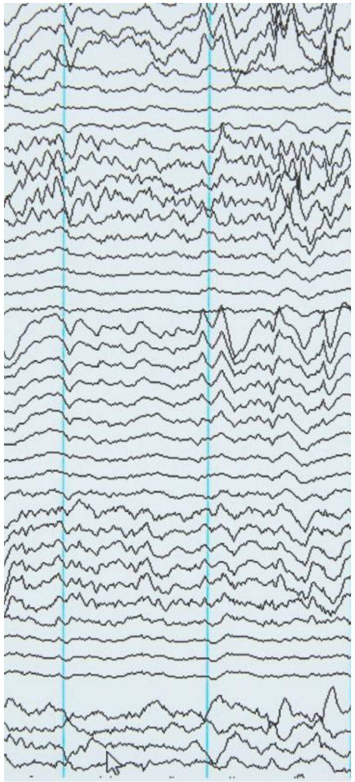
mainly used in connection with applying known machine learning and statistical methods on text when addressing tasks such as Document Clustering or Document Classification

Pattern Discovery

- Analyzes concept co-occurrence patterns across documents to discover relationships.

  Ex:-

  - *Classification*
  - *Clustering*
  - *Trend Analysis*
  - *Anomaly Detection*



Use of Text Mining

- Benefits of text mining are obvious especially in text-rich data environments such as
  - *law (court orders), academic research (research articles), finance (quarterly reports),*
  - *medicine (discharge summaries), biology (molecular interactions), technology (patent files), marketing (customer comments), etc.*

Use of Text Mining

- Electronic communication records (e.g., Email)
  - Spam filtering
  - Email prioritization and categorization
  - Automatic response generation

Today's Example

- Classification of free-text documents is a common task in the field of text mining.
- It is used to categorize documents

  i.e. assign pre-defined topics,

  or it can be used for sentiment analysis.

Today's Example

- Today we want to construct a workflow that

1. *reads and preprocesses text documents,*
2. *transforms them into a numerical representation*
3. *builds a predictive model to assign pre-defined labels to documents.*

Today's Example

- Additional tasks:

  Sentiment analysis

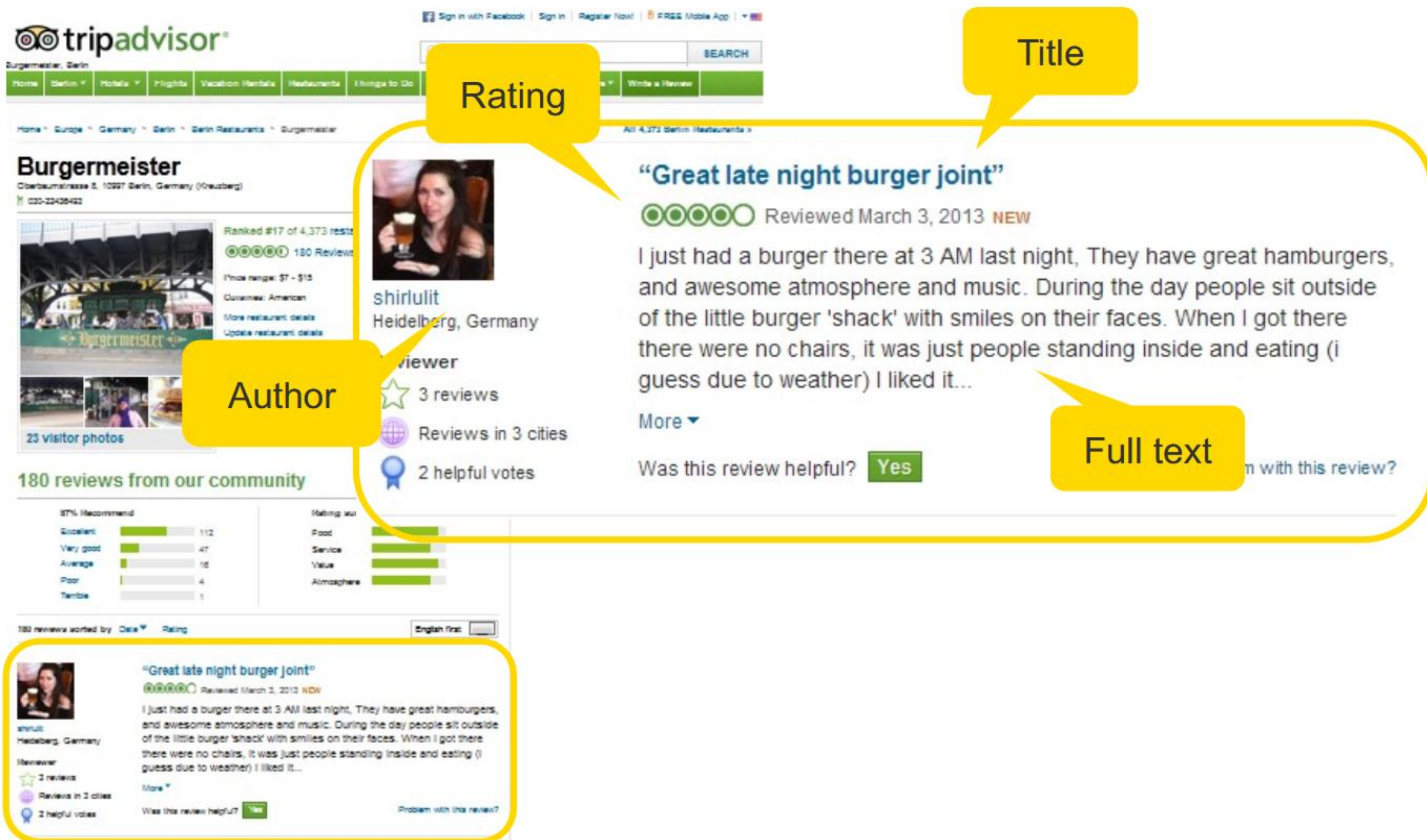
  Visualization of documents

  Document clustering

  Topic Modeling

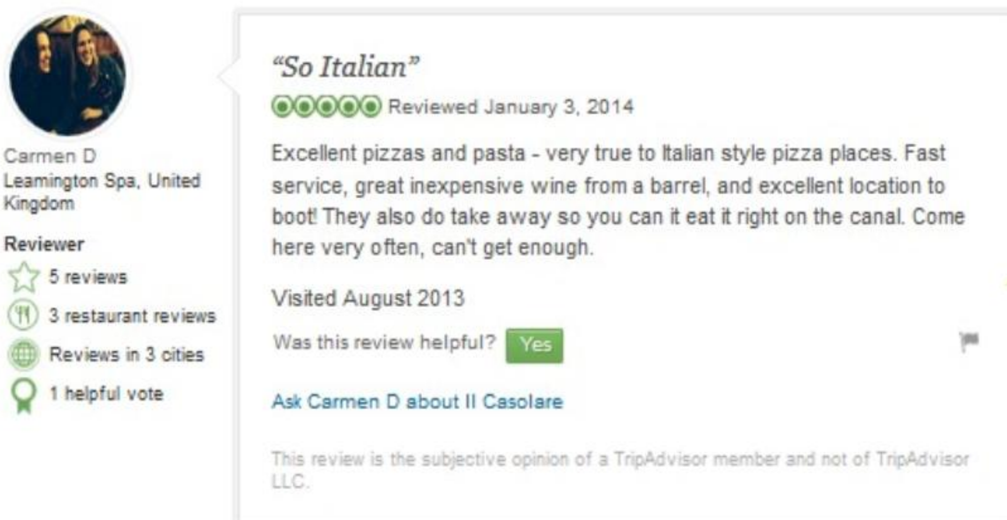
  Text Summarization

Today's Example



Today's Example

- Build a classifier to distinguish between reviews about Italian or Chinese restaurants.



Review about an Italian or a Chinese restaurant?

Examples of Text Mining Applications

- Product review sentiment analysis
- Automatic email routing
- Detecting fake news
- Market trend monitoring
- Customer service automation
- Medical text analysis in healthcare
- Social media monitoring
- Risk & fraud detection

Common Challenges

- Data noise (slang, typos, emojis)
- Ambiguity in language
- Domain-specific terms
- Multilingual and code-mixed data
- High computational requirements
- Large dataset processing

Conclusion

Text mining systems transform unstructured data into structured formats

enabling advanced analysis and discovery of patterns and trends.

References

Feldman, Ronen, and James Sanger. *The text mining handbook: advanced approaches in analyzing unstructured data*. Cambridge university press, 2007.